



ARTIFICIAL
NEURAL NETWORK

FINAL PROJECT

SEMINARIO DE SOLUCION DE
PROBLEMAS DE SISTEMAS BASADOS EN
CONOCIMIENTOS

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FAITH TEAM



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Diosito

INTRODUCTION

Algorithms

- Logistic Regresión
- K Nearest Neighbors
- Neural Network

Rationale

- easy implementation
- Good for modeling probabilities
- Binary dependent variable
- Classification Problems



EVALUATION METRICS

$$\text{Accuracy} = \frac{\text{Correct predictions}}{\text{All predictions}}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Sensitivity} = \frac{\text{True positive}}{\text{True positive} + \text{false negative}}$$

$$F_1 = 2 * \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}}$$



Logistic Regression

Dataset: Employee

Label Encoding (Education, Joining Year, City, Gender, EverBenched)

StandarScaler

Description: evaluation with employee data to know if they leave or not

	A	B	C	D	E	F	G	H	I
1	Education	JoiningYear	City	PaymentTier	Age	Gender	EverBenched	ExperienceIn	LeaveOrNot
2	Bachelors	2017	Bangalore		3	34 Male	No	0	0
3	Bachelors	2013	Pune		1	28 Female	No	3	1
4	Bachelors	2014	New Delhi		3	38 Female	No	2	0
5	Masters	2016	Bangalore		3	27 Male	No	5	1
6	Masters	2017	Pune		3	24 Male	Yes	2	1
7	Bachelors	2016	Bangalore		3	22 Male	No	0	0
8	Bachelors	2015	New Delhi		3	38 Male	No	0	0
9	Bachelors	2016	Bangalore		3	34 Female	No	2	1
10	Bachelors	2016	Pune		3	23 Male	No	1	0
11	Masters	2017	New Delhi		2	37 Male	No	2	0
12	Masters	2012	Bangalore		3	27 Male	No	5	1
13	Bachelors	2016	Pune		3	34 Male	No	3	0
14	Bachelors	2018	Pune		3	32 Male	Yes	5	1

Logistic Regression

```
def train():  
    for i in range(0, len(X_train)):  
        yp_train[i] = sigmoide(calc(X_train[i, :], w))  
        #print(yp_train[i])
```

```
def calc(a, b):  
    z = 0  
    for j in range(len(a)):  
        z = z + (a[j] * b[j+1])  
    return z + b[0]
```

```
def sigmoide(c):  
    return 1 / (1 + np.exp(-c))
```

```
#Actualizacion de pesos  
def actualizacion(pesos):  
    for j in range(0, len(pesos)):  
        w[j] = w[j] - (alpha * pesos[j])
```

```
yp_test = np.zeros(len(y_test))  
def test():  
    #print(w)  
    for i in range(0, len(X_test)):  
        yp_test[i] = sigmoide(calc(X_test[i, :], w))  
        #print(yp_test[i], y_test[i])
```

```
def cross_entropy(yi, yj):  
    m = len(yi)  
    epsilon = 1e-15  
    b = 0  
    for i in range(m):  
        b = b + (yi[i] * np.log(yj[i] + epsilon) + (1 - yi[i]) * np.log(1 - yj[i] + epsilon))  
    z = (-1/m) * b  
    return z
```

RESULTS

Accuracy: 71.21374865735768
Precision: 63.730569948186535
Sensitivity: 38.31775700934579
F1 Score: 47.859922178988334
TP: 123
FP: 70
TN: 540
FN: 198

Pesos Iniciales:

[4.63167696 3.64230169 6.46324639 8.41639837 4.42423753 1.75279089
4.28105508 6.08904287 3.44863197]

Pesos Finales:

[-0.73192061 0.04902807 0.35923614 0.31415167 -0.20616743 -0.07948001
-0.37332837 0.16850466 -0.06894864]

	Real		
Prediction		Positive (1)	Negative (0)
	Positive (1)	TP: 123	FP: 70
	Negative (0)	F.N: 198	TN: 540

Error entrenamiento: 6.384278187777074
Error entrenamiento: 5.4183291648141365
Error entrenamiento: 4.49653345281625
Error entrenamiento: 3.6414847178161764
Error entrenamiento: 2.8708575561418073
Error entrenamiento: 2.2026262303416537
Error entrenamiento: 1.6502781558347115
Error entrenamiento: 1.225615408069671
Error entrenamiento: 0.9234808866535807
Error entrenamiento: 0.7281958076257157
Error entrenamiento: 0.6244522508615978
Error entrenamiento: 0.588859530119828
Error entrenamiento: 0.5832974924163677
Error entrenamiento: 0.5829387830500712
Error entrenamiento: 0.5829248259957225
Error entrenamiento: 0.5829243523634462
Error entrenamiento: 0.5829243328303656
Error entrenamiento: 0.5829243291169324

Logistic Regression

Dataset: Heart

StandardScaler

Description:

age
sex
chest pain type (4 values)
resting blood pressure
serum cholestoral in mg/dl
fasting blood sugar > 120 mg/dl
resting electrocardiographic results (values 0,1,2)
maximum heart rate achieved
exercise induced angina
oldpeak = ST depression induced by exercise relative to rest
the slope of the peak exercise ST segment
number of major vessels (0-3) colored by flourosopy
thal: 0 = normal; 1 = fixed defect; 2 = reversable defect

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	age	sex	cp	trtbps	chol	fbs	restecg	thalachh	exng	oldpeak	slp	caa	thall	output
2	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
3	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
4	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
5	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
6	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1
7	57	1	0	140	192	0	1	148	0	0.4	1	0	1	1
8	56	0	1	140	294	0	0	153	0	1.3	1	0	2	1
9	44	1	1	120	263	0	1	173	0	0	2	0	3	1
10	52	1	2	172	199	1	1	162	0	0.5	2	0	3	1
11	57	1	2	150	168	0	1	174	0	1.6	2	0	2	1
12	54	1	0	140	239	0	1	160	0	1.2	2	0	2	1
13	48	0	2	130	275	0	1	139	0	0.2	2	0	2	1
14	49	1	1	130	266	0	1	171	0	0.6	2	0	2	1
15	64	1	3	110	211	0	0	144	1	1.8	1	0	2	1
16	58	0	3	150	283	1	0	162	0	1	2	0	2	1
17	50	0	2	120	219	0	1	158	0	1.6	1	0	2	1

The "target" field refers to the presence of heart disease in the patient. It is integer valued 0 = no disease and 1 = disease.



RESULTS

Accuracy: 78.68852459016394
Precision: 78.78787878787878
Sensitivity: 81.25
F1 Score: 80.0
TP: 26
FP: 7
TN: 22
FN: 6

Pesos Iniciales:

```
[9.55525689 9.63894302 4.9837471 6.57191305 5.66911704 2.26514974  
9.07063265 2.82953361 0.65144626 3.37500497 8.87593401 4.6187245  
5.55962145 9.11755432]
```

Pesos finales:

```
[ 0.123973 2.21182598 -6.52497167 7.35456085 -2.57029101  
-2.94362711 3.77141885 1.95726328 2.86853239 -5.4755401  
-1.77728031 6.49223785 -20.99071098 9.11755432]
```

```
Error entrenamiento: 8.695453658891442  
Error entrenamiento: 3.765671835086858  
Error entrenamiento: 2.0645609888980756  
Error entrenamiento: 1.4700379389743659  
Error entrenamiento: 1.0960679090973913  
Error entrenamiento: 0.8782188611176605  
Error entrenamiento: 0.7476806142695775  
Error entrenamiento: 0.6870831411498327  
Error entrenamiento: 0.660460473516544  
Error entrenamiento: 0.6510460320934688  
Error entrenamiento: 0.6542257051973244  
Error entrenamiento: 0.6698358658959528
```

	Real		
Prediction		Positive (1)	Negative (0)
	Positive (1)	TP: 26	FP: 7
	Negative (0)	F.N: 6	TN: 22

Logistic Regression

Dataset: Zoo One Hot Encoding and Estándar Scaler

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	animal_name	hair	feathers	eggs	milk	airborne	aquatic	predator	toothed	backbone	breathes	venomous	fins	legs	tail	domestic	catsize	class_type
2	aardvark		1	0	0	1	0	0	1	1	1	1	0	0	4	0	0	1
3	antelope		1	0	0	1	0	0	0	1	1	1	0	0	4	1	0	1
4	bass		0	0	1	0	0	1	1	1	1	0	0	1	0	1	0	4
5	bear		1	0	0	1	0	0	1	1	1	1	0	0	4	0	0	1
6	boar		1	0	0	1	0	0	1	1	1	1	0	0	4	1	0	1
7	buffalo		1	0	0	1	0	0	0	1	1	1	0	0	4	1	0	1
8	calf		1	0	0	1	0	0	0	1	1	1	0	0	4	1	1	1
9	carp		0	0	1	0	0	1	0	1	1	0	0	1	0	1	0	4
10	catfish		0	0	1	0	0	1	1	1	1	0	0	1	0	1	0	4
11	cavy		1	0	0	1	0	0	0	1	1	1	0	0	4	0	1	1
12	cheetah		1	0	0	1	0	0	1	1	1	1	0	0	4	1	0	1
13	chicken		0	1	1	0	1	0	0	1	1	1	0	0	2	1	1	2
14	chub		0	0	1	0	0	1	1	1	1	0	0	1	0	1	0	4
15	clam		0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	7
16	crab		0	0	1	0	0											
17	crayfish		0	0	1	0	0											
18	crow		0	1	1	0	1											
19	deer		1	0	0	1	0											
20	dogfish		0	0	1	0	0											
21	dolphin		0	0	0	1	0											
22	dove		0	1	1	0	1											

This dataset consists of 101 animals from a zoo. There are 16 variables with various traits to describe the animals.

The 7 Class Types are: Mammal, Bird, Reptile, Fish, Amphibian, Bug and Invertebrate

The purpose for this dataset is to be able to predict the classification of the animals, based upon the variables.

	A	B	C	D	E	F	G	H	I
1	Class_Numb	Number_Of	Class_Type	Animal_Names					
2	1	41	Mammal	aardvark, antelope, bear, boar, buffalo, calf, cavy, cheetah, deer, dolphin, ele					
3	2	20	Bird	chicken, crow, dove, duck, flamingo, gull, hawk, kiwi, lark, ostrich, parakeet, p					
4	3	5	Reptile	pitviper, seasnake, slowworm, tortoise, tuatara					
5	4	13	Fish	bass, carp, catfish, chub, dogfish, haddock, herring, pike, piranha, seahorse, s					
6	5	4	Amphibian	frog, frog, newt, toad					
7	6	8	Bug	flea, gnat, honeybee, housefly, ladybird, moth, termite, wasp					
8	7	10	Invertebrate	clam, crab, crayfish, lobster, octopus, scorpion, seawasp, slug, starfish, worm					

RESULTS

```
#one hot encoder
```

```
encoder = OneHotEncoder(sparse=False)
y_resaped = y.reshape(-1, 1)
y = encoder.fit_transform(y_resaped)
```

```
Accuracy: 95.23809523809523
Precision: 95.23809523809523
Sensitivity: 95.23809523809523
F1 Score: 95.23809523809523
TP: 19
FP: 1
TN: 105
FN: 1
```

	Real		
	Prediction	Positive (1)	Negative (0)
		TP: 19	FP: 1
		F.N: 1	TN: 105

```
Error: 19.131681962532987
Error: 4.2845465421787905
Error: 2.6325202365488987
Error: 1.845844351848585
Error: 1.4177163173436322
Error: 1.1018806456011405
Error: 0.8163859184614009
Error: 0.5219151940677001
Error: 0.2898648481165745
Error: 0.1245639178323269
Error: 0.04413061740701626
Error: 0.026663938729950248
Error: 0.018720150966083142
Error: 0.015871062505942037
Error: 0.013906612534817647
Error: 0.012436875122604132
Error: 0.011288498357665253
Error: 0.010362692362575867
Error: 0.009598224611849905
Error: 0.008954790762305353
Error: 0.008404651488386426
Error: 0.007928039181109348
Error: 0.007510465238132144
Error: 0.007141059187134651
Error: 0.006811501568824324
Error: 0.0065153149207111665
Error: 0.00624737926172164
Error: 0.006003592969658222
Error: 0.005780630459951511
Error: 0.005575765868877298
```

Prediction vs Real

```
[1 0 0 0 0 0 0] [1 0 0 0 0 0 0]
[1 0 0 0 0 0 0] [1 0 0 0 0 0 0]
[1 0 0 0 0 0 0] [1 0 0 0 0 0 0]
[1 0 0 0 0 0 0] [1 0 0 0 0 0 0]
[1 0 0 0 0 0 0] [1 0 0 0 0 0 0]
[0 0 0 0 0 1 0] [0 0 0 0 0 1 0]
[1 0 0 0 0 0 0] [1 0 0 0 0 0 0]
[1 0 0 0 0 0 0] [1 0 0 0 0 0 0]
[1 0 0 0 0 0 0] [1 0 0 0 0 0 0]
[1 0 0 0 0 0 0] [1 0 0 0 0 0 0]
[1 0 0 0 0 0 0] [1 0 0 0 0 0 0]
[0 0 0 1 0 0 0] [0 0 0 1 0 0 0]
[0 0 0 0 0 1 0] [0 0 0 0 0 1 0]
[0 0 0 0 0 1 0] [0 0 0 0 0 1 0]
[0 1 0 0 0 0 0] [0 1 0 0 0 0 0]
[0 0 0 0 0 0 1] [0 0 0 0 0 0 1]
[1 0 0 0 0 0 0] [1 0 0 0 0 0 0]
[0 1 0 0 0 0 0] [1 0 0 0 0 0 0]
[0 1 0 0 0 0 0] [0 1 0 0 0 0 0]
[0 0 0 1 0 0 0] [0 0 0 1 0 0 0]
[1 0 0 0 0 0 0] [1 0 0 0 0 0 0]
[0 0 0 0 0 0 1] [0 0 1 0 0 0 0]
```

Pesos Iniciales:

```
[[ 7.34713215  4.46227592 10.30666274  6.35377003  7.31962983  5.0745418
  1.3223885   4.49717203  1.86304942  0.81689981  8.54022679  1.8693512
  2.90707226  3.71433752  5.65417355  8.27339532]
 [ 3.70633728  0.58015287 10.20282906  9.83962225  3.70159883  9.97707733
 10.62619925  7.23051381  7.66710171  9.16453447  0.67548465  7.18509529
  8.29193697  0.09406605  6.25476937  6.06551939]
 [ 5.89943227  6.86995627  5.74540451  1.43999617  3.94019346  6.04240077
  5.98706796  2.1950372  4.2103366  0.93367936  0.58614319  6.1740023
 10.10081771  1.78329134  2.99750526  3.16704243]
 [ 4.04950436  1.51396756  2.21369197  5.11185738  7.54332726  0.45692615
  0.79137878  7.98471871  6.44124635  1.26214139  2.91593711  3.47810878
  1.15805023  4.70389595  2.41211946  2.08062961]
 [ 3.60036971  4.10925679 10.29801751  5.2135523  8.59675768  0.86562654
  6.26542211  5.10488009  0.35817465  5.95478826  3.21801088  6.5707358
  1.36267983 10.35483297  6.67178663  9.83348501]
 [ 8.37113789  4.13235515  6.76972656  1.76586437  2.10625449  2.32379528
  6.95520611  0.38900321  0.66857647  3.57732742  7.17204479 10.58329565
  1.90105892  1.40755117  6.39513164  4.60070077]
 [ 6.58292874  5.75537336  9.97383921  0.9272653 10.30564558  4.66128697
  0.90513404 10.31124218  4.03618515  8.22512672  0.25998405  3.77030251
  6.08969328  1.53981429  6.05152741  7.81078705]]
```

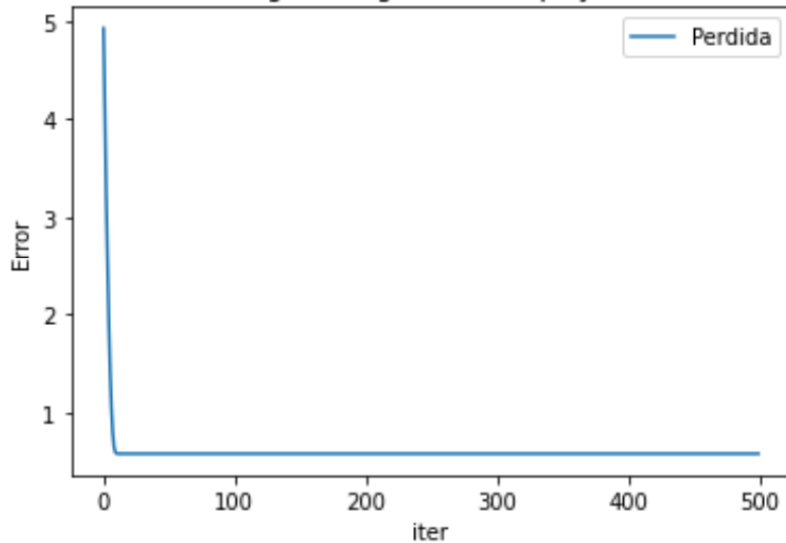
Pesos Finales:

```
[[11.28748291  2.99397676  5.71221157 11.15381799  6.11360204  4.44132591
  1.50344416  7.01030622  3.09539045  2.51857234  6.10145462  2.24579667
  3.24016261  4.09864509  6.50436907 10.58723979]
 [ 0.86038235  8.22240987 13.0628534  6.7740864  8.84803909  6.68659293
  7.15009177  1.2632706  7.79381065  9.30971673  0.46048419  6.27467527
  5.82909005  2.94523466  7.03954848  5.34950469]
 [ 4.98022432  4.70053133  6.91731099  0.54518983  2.65384768  2.52092094
  5.89355358  4.64751016  7.46602972  3.09944068  3.6465645  5.62610619
  6.14011195  5.12653456  2.52370194  3.49239175]
 [ 2.57462005  1.68670171  3.77764351  3.54790584  7.47954359  3.3779172
  1.80532455  7.36853358  5.90740307 -1.78383427  3.45800716  6.59217329
  0.60893511  4.27674596  2.81632965  2.75476746]
 [ 2.78117156  1.05282453 11.60341286  3.90832289  6.14477612  7.03974752
  6.34093724  8.4332415  0.8687134  6.62120281  3.47818849  5.33954722
  5.63032131  4.40974116  5.58109658  6.9343372 ]
 [11.27065396  5.00615542  6.7631828  3.76537216  6.08666747  1.58266378
  5.39655666  1.58207012  2.14965469  6.11740641  2.07450462  9.39648299
  6.36651813  0.57391299  7.26522383  4.89743166]
 [ 5.80230725  3.76073829  7.67355643  0.9572327  6.18693114  3.75248655
  4.76288878  7.40763505 -2.03633162  4.05199272  4.14862787  4.1561099
  3.99617003  2.16697489  4.70674377  7.81588704]]
```

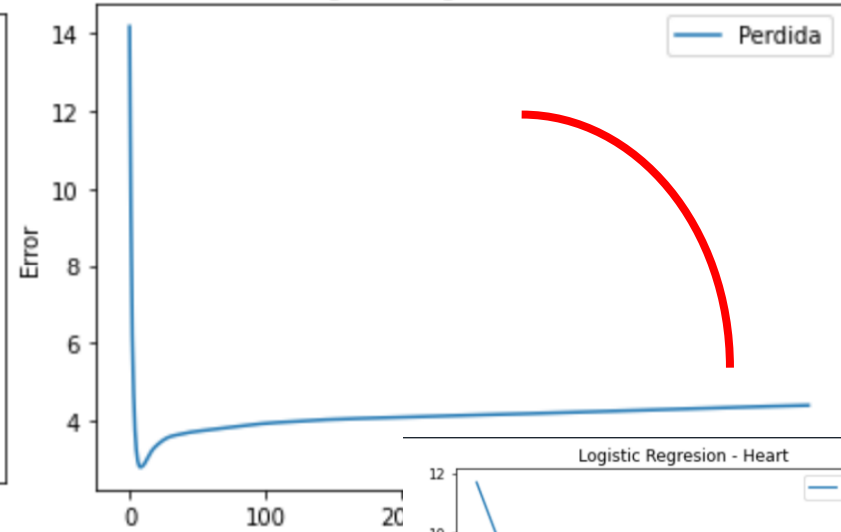
RESULTS COMPARISON



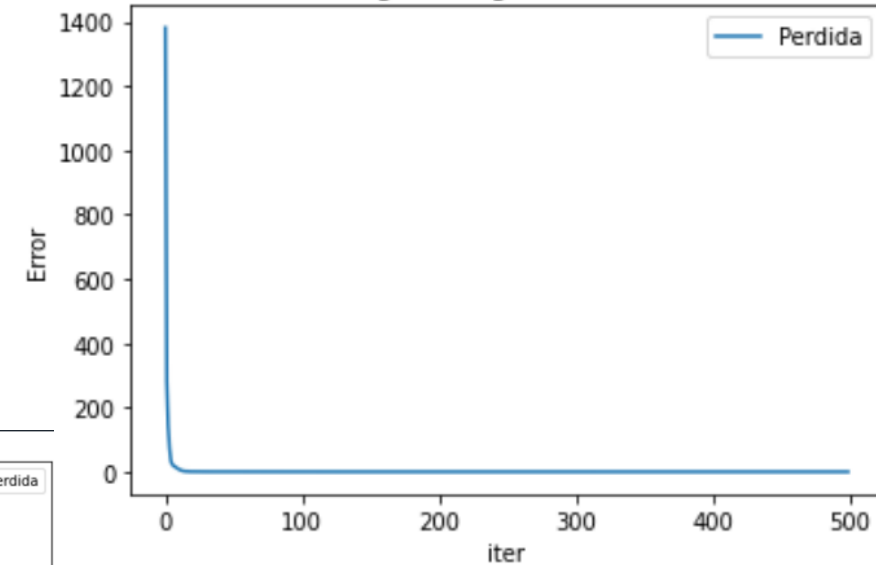
Logistic Regression - Employee



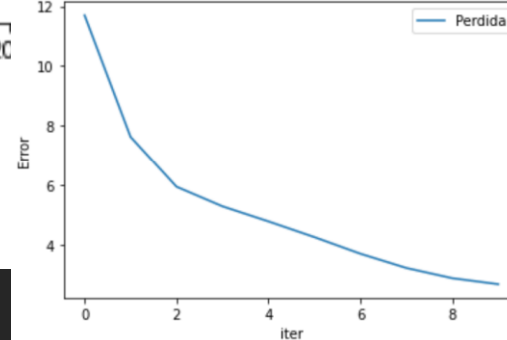
Logistic Regression - Heart



Logistic Regression - Zoo



Logistic Regression - Heart



K Nearest Neighbors

Dataset: Employee

Label Encoding (Education, Joining Year, City, Gender, EverBenched)

StandardScaler

Description: evaluation with employee data to know if they leave or not

	A	B	C	D	E	F	G	H	I
1	Education	JoiningYear	City	PaymentTier	Age	Gender	EverBenched	ExperienceIn	LeaveOrNot
2	Bachelors	2017	Bangalore		3	34 Male	No	0	0
3	Bachelors	2013	Pune		1	28 Female	No	3	1
4	Bachelors	2014	New Delhi		3	38 Female	No	2	0
5	Masters	2016	Bangalore		3	27 Male	No	5	1
6	Masters	2017	Pune		3	24 Male	Yes	2	1
7	Bachelors	2016	Bangalore		3	22 Male	No	0	0
8	Bachelors	2015	New Delhi		3	38 Male	No	0	0
9	Bachelors	2016	Bangalore		3	34 Female	No	2	1
10	Bachelors	2016	Pune		3	23 Male	No	1	0
11	Masters	2017	New Delhi		2	37 Male	No	2	0
12	Masters	2012	Bangalore		3	27 Male	No	5	1
13	Bachelors	2016	Pune		3	34 Male	No	3	0
14	Bachelors	2018	Pune		3	32 Male	Yes	5	1

K = 3

RESULTS

```
def etiquetar(yi, pos):
    orden = np.argsort(yi[:, 0])
    yiord = yi[orden]
    v1 = 0
    v0 = 0
    for q in range(k):
        if(yiord[k, 1] == 1):
            v1 = v1 + 1
        if(yiord[k, 1] == 0):
            v0 = v0 + 1

    if(v1 > v0):
        yp_test[pos] = 1
    if(v0 > v1):
        yp_test[pos] = 0

def eucl(x1, x2, pos):
    d = np.zeros((len(x2), 2))

    for j in range(len(x2)):
        r = 0
        for k in range(len(x1)):
            r = r + ((x1[k] - x2[j, k])**2)
        d[j, 0] = np.sqrt(r)
        d[j, 1] = y_train[pos]

    print(d)
    etiquetar(d, pos)
```

Accuracy: 74.11385606874329
Precision: 63.0718954248366
Sensitivity: 60.12461059190031
F1 Score: 61.56299840510366
TP: 193
FP: 113
TN: 497
FN: 128

```
...
[5.48746866 0.      ]
[5.16198067 0.      ]
[5.26222659 1.      ]
[3.12646422 1.      ]
[0.55026365 0.      ]
[4.24724334 0.      ]
...
[4.08693985 0.      ]
[3.67840031 0.      ]
[1.71199072 1.      ]
[5.41578234 1.      ]
[5.12488002 0.      ]
[4.34662507 0.      ]
...
[5.61480277 0.      ]
[5.57436488 0.      ]
[4.28746606 1.      ]
[3.77122904 1.      ]
[2.62822511 0.      ]
[4.91840315 0.      ]
...
```

	Real		
Prediction		Positive (1)	Negative (0)
	Positive (1)	TP: 193	FP: 113
	Negative (0)	F.N: 128	TN: 497

K Nearest Neighbors

Dataset: Heart

StandardScaler

Description: Description:

age

sex

chest pain type (4 values)

resting blood pressure

serum cholestoral in mg/dl

fasting blood sugar > 120 mg/dl

resting electrocardiographic results (values 0,1,2)

maximum heart rate achieved

exercise induced angina

oldpeak = ST depression induced by exercise relative to rest

the slope of the peak exercise ST segment

number of major vessels (0-3) colored by flourosopy

thal: 0 = normal; 1 = fixed defect; 2 = reversable defect

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	age	sex	cp	trtbps	chol	fbs	restecg	thalachh	exng	oldpeak	slp	caa	thall	output
2	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
3	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
4	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
5	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
6	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1
7	57	1	0	140	192	0	1	148	0	0.4	1	0	1	1
8	56	0	1	140	294	0	0	153	0	1.3	1	0	2	1
9	44	1	1	120	263	0	1	173	0	0	2	0	3	1
10	52	1	2	172	199	1	1	162	0	0.5	2	0	3	1
11	57	1	2	150	168	0	1	174	0	1.6	2	0	2	1
12	54	1	0	140	239	0	1	160	0	1.2	2	0	2	1
13	48	0	2	130	275	0	1	139	0	0.2	2	0	2	1
14	49	1	1	130	266	0	1	171	0	0.6	2	0	2	1
15	64	1	3	110	211	0	0	144	1	1.8	1	0	2	1
16	58	0	3	150	283	1	0	162	0	1	2	0	2	1
17	50	0	2	120	219	0	1	158	0	1.6	1	0	2	1

RESULTS

Prediction vs Real

1 1
0 0
1 0
1 1
0 0
1 1
1 1
1 1
1 1
1 0
1 1
1 1
0 0
1 1
0 0
1 1
1 0
0 0
0 0
0 0
1 1
1 0
0 0
0 0
0 1
1 1
0 0
1 1
0 1
0 0

Accuracy: 74.72527472527473
Precision: 75.47169811320755
Sensitivity: 80.0
F1 Score: 77.66990291262137
TP: 40
FP: 13
TN: 28
FN: 10

[6.47207026 0.]
[6.03565056 0.]
[6.15812535 1.]
[5.17574224 1.]
[4.91439746 0.]
[7.05481148 0.]
[6.45757589 0.]
[5.43874113 1.]
[4.58218014 1.]
[3.45647842 1.]
[7.65703579 1.]
[6.68943976 1.]
[7.02798111 0.]
[6.5298768 1.]
[5.52769065 0.]
[4.99172259 1.]
[5.19298874 0.]
[6.21000068 1.]
[6.10369095 1.]
[6.70963175 0.]

	Real		
Prediction		Positive (1)	Negative (0)
	Positive (1)	TP: 40	FP: 13
	Negative (0)	F.N: 28	TN: 10

K Nearest Neighbors

Dataset: Zoo

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	animal_name	hair	feathers	eggs	milk	airborne	aquatic	predator	toothed	backbone	breathes	venomous	fins	legs	tail	domestic	catsize	class_type
2	aardvark		1	0	0	1	0	0	1	1	1	1	0	0	4	0	0	1
3	antelope		1	0	0	1	0	0	0	1	1	1	0	0	4	1	0	1
4	bass		0	0	1	0	0	1	1	1	1	0	0	1	0	1	0	4
5	bear		1	0	0	1	0	0	1	1	1	1	0	0	4	0	0	1
6	boar		1	0	0	1	0	0	1	1	1	1	0	0	4	1	0	1
7	buffalo		1	0	0	1	0	0	0	1	1	1	0	0	4	1	0	1
8	calf		1	0	0	1	0	0	0	1	1	1	0	0	4	1	1	1
9	carp		0	0	1	0	0	1	0	1	1	0	0	1	0	1	0	4
10	catfish		0	0	1	0	0	1	1	1	1	0	0	1	0	1	0	4
11	cavy		1	0	0	1	0	0	0	1	1	1	0	0	4	0	1	1
12	cheetah		1	0	0	1	0	0	1	1	1	1	0	0	4	1	0	1
13	chicken		0	1	1	0	1	0	0	1	1	1	0	0	2	1	1	2
14	chub		0	0	1	0	0	1	1	1	1	0	0	1	0	1	0	4
15	clam		0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	7
16	crab		0	0	1	0	0											
17	crayfish		0	0	1	0	0											
18	crow		0	1	1	0	1											
19	deer		1	0	0	1	0											
20	dogfish		0	0	1	0	0											
21	dolphin		0	0	0	1	0											
22	dove		0	1	1	0	1											

One Hot Encoding
Estándar Scaler

	A	B	C	D	E	F	G	H	I
1	Class_Numb	Number_Of	Class_Type	Animal_Names					
2	1	41	Mammal	aardvark, antelope, bear, boar, buffalo, calf, cavy, cheetah, deer, dolphin, ele					
3	2	20	Bird	chicken, crow, dove, duck, flamingo, gull, hawk, kiwi, lark, ostrich, parakeet, p					
4	3	5	Reptile	pitvip, seasnake, slowworm, tortoise, tuatara					
5	4	13	Fish	bass, carp, catfish, chub, dogfish, haddock, herring, pike, piranha, seahorse, s					
6	5	4	Amphibian	frog, frog, newt, toad					
7	6	8	Bug	flea, gnat, honeybee, housefly, ladybird, moth, termite, wasp					
8	7	10	Invertebrate	clam, crab, crayfish, lobster, octopus, scorpion, seawasp, slug, starfish, worm					

RESULTS

Accuracy: 90.47619047619048
Categoria: 1
Precision: 91.66666666666666
Sensitivity: 100.0
F1 Score Cateogira: 95.65217391304347
Categoria: 2
Precision: 100.0
Sensitivity: 100.0
F1 Score Cateogira: 100.0
Categoria: 3
Precision: 0.0
Sensitivity: 0.0
F1 Score Cateogira: 0.0
Categoria: 4
Precision: 100.0
Sensitivity: 50.0
F1 Score Cateogira: 66.66666666666667
Categoria: 5
Precision: 0.0
Sensitivity: 0.0
F1 Score Cateogira: 0.0
Categoria: 6
Precision: 100.0
Sensitivity: 100.0
F1 Score Cateogira: 100.0
Categoria: 7
Precision: 100.0
Sensitivity: 100.0
F1 Score Cateogira: 100.0

Verdaderos Positivos Categoria: 1 : 11.0
Falsos Negativos Categoria: 1 : 1.0
Verdaderos Negativos Categoria: 1 : 9.0
Falsos Positivos Categoria: 1 : 0.0
Verdaderos Positivos Categoria: 2 : 2.0
Falsos Negativos Categoria: 2 : 0.0
Verdaderos Negativos Categoria: 2 : 19.0
Falsos Positivos Categoria: 2 : 0.0
Verdaderos Positivos Categoria: 3 : 0.0
Falsos Negativos Categoria: 3 : 1.0
Verdaderos Negativos Categoria: 3 : 20.0
Falsos Positivos Categoria: 3 : 0.0
Verdaderos Positivos Categoria: 4 : 2.0
Falsos Negativos Categoria: 4 : 0.0
Verdaderos Negativos Categoria: 4 : 17.0
Falsos Positivos Categoria: 4 : 2.0
Verdaderos Positivos Categoria: 5 : 0.0
Falsos Negativos Categoria: 5 : 0.0
Verdaderos Negativos Categoria: 5 : 21.0
Falsos Positivos Categoria: 5 : 0.0
Verdaderos Positivos Categoria: 6 : 3.0
Falsos Negativos Categoria: 6 : 0.0
Verdaderos Negativos Categoria: 6 : 18.0
Falsos Positivos Categoria: 6 : 0.0
Verdaderos Positivos Categoria: 7 : 1.0
Falsos Negativos Categoria: 7 : 0.0
Verdaderos Negativos Categoria: 7 : 20.0
Falsos Positivos Categoria: 7 : 0.0

Accuracy:
90.47

Real Clase 1			
Prediction			
	Positive (1)		Negative (0)
	Positive (1)	TP: 11	FP: 0
	Negative (0)	F.N: 1	TN: 9



Neural Network

Dataset: Employee

Label Encoding (Education, Joining Year, City, Gender, EverBenched)

StandarScaler

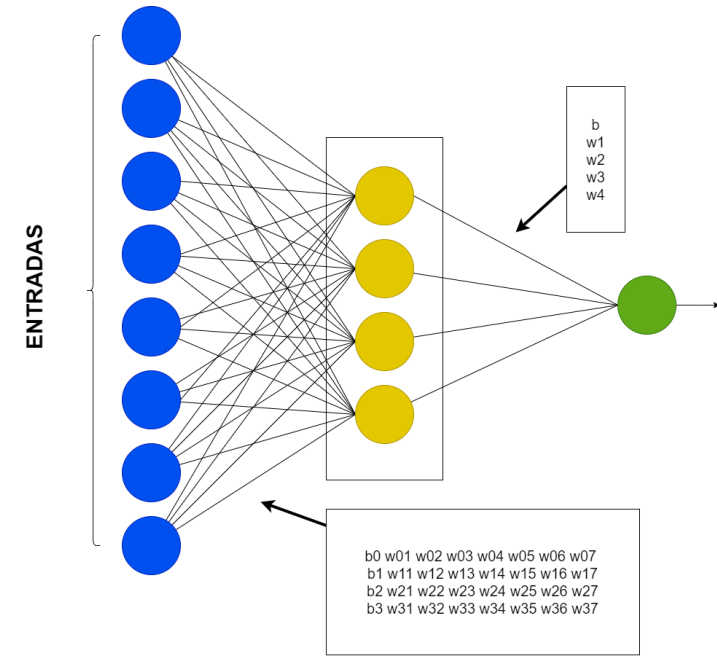
Description: evaluation with employee data to know if they leave or not

	A	B	C	D	E	F	G	H	I
1	Education	JoiningYear	City	PaymentTier	Age	Gender	EverBenched	ExperienceIn	LeaveOrNot
2	Bachelors	2017	Bangalore		3	34 Male	No	0	0
3	Bachelors	2013	Pune		1	28 Female	No	3	1
4	Bachelors	2014	New Delhi		3	38 Female	No	2	0
5	Masters	2016	Bangalore		3	27 Male	No	5	1
6	Masters	2017	Pune		3	24 Male	Yes	2	1
7	Bachelors	2016	Bangalore		3	22 Male	No	0	0
8	Bachelors	2015	New Delhi		3	38 Male	No	0	0
9	Bachelors	2016	Bangalore		3	34 Female	No	2	1
10	Bachelors	2016	Pune		3	23 Male	No	1	0
11	Masters	2017	New Delhi		2	37 Male	No	2	0
12	Masters	2012	Bangalore		3	27 Male	No	5	1
13	Bachelors	2016	Pune		3	34 Male	No	3	0
14	Bachelors	2018	Pune		3	32 Male	Yes	5	1

$$\frac{\partial C}{\partial w^L} = \frac{\partial C}{\partial a^L} \cdot \frac{\partial a^L}{\partial z^L} \cdot \frac{\partial z^L}{\partial w^L}$$

$$\frac{\partial C}{\partial b^L} = \frac{\partial C}{\partial a^L} \cdot \frac{\partial a^L}{\partial z^L} \cdot \frac{\partial z^L}{\partial b^L}$$

$$\frac{\partial a^L}{\partial z^L} = a^L(z^L) \cdot (1 - a^L(z^L)) \quad \frac{\partial C}{\partial a_j^L} = (a_j^L - y_j) \quad \frac{\partial z^L}{\partial b^L} = 1 \quad \frac{\partial z^L}{\partial w^L} = a_i^{L-1}$$



OPERATION

$$\delta^L = \frac{\partial C}{\partial w^{L-1}} = \frac{\partial C}{\partial a^L} \cdot \frac{\partial a^L}{\partial z^L} \cdot \frac{\partial z^L}{\partial a^{L-1}} \cdot \frac{\partial a^{L-1}}{\partial z^{L-1}} \cdot \frac{\partial z^{L-1}}{\partial w^{L-1}}$$

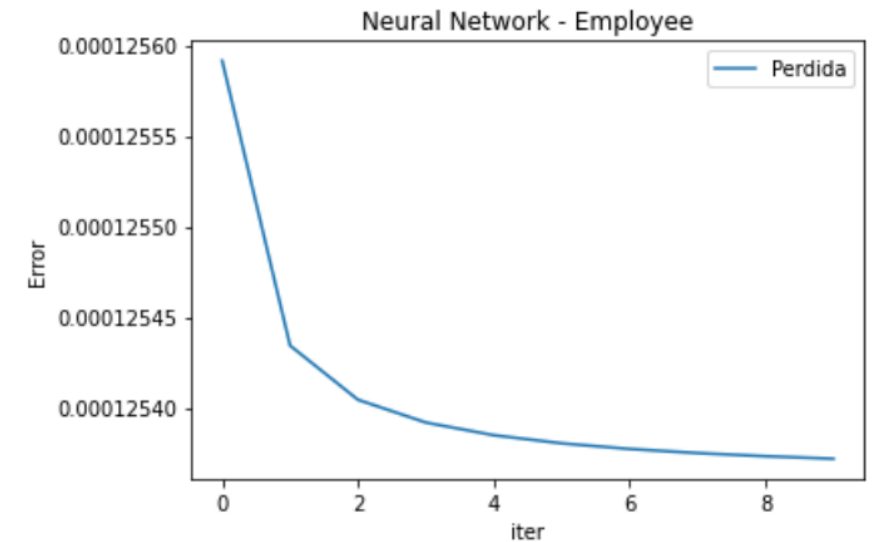
$$\frac{\partial C}{\partial b^{L-1}} = \frac{\partial C}{\partial a^L} \cdot \frac{\partial a^L}{\partial z^L} \cdot \frac{\partial z^L}{\partial a^{L-1}} \cdot \frac{\partial a^{L-1}}{\partial z^{L-1}} \cdot \frac{\partial z^{L-1}}{\partial b^{L-1}}$$

APLICAMOS LA CHAIN RULE A ESTA COMPOSICION

$$C(a^L(W^L a^{L-1}(W^{L-1} a^{L-2} + b^{L-1}) + b^L))$$

RESULTS

```
CUCEI/Sem Sistemas Basados/Proyecto')
Accuracy: 64.55424274973147
Precision: 45.714285714285715
Sensitivity: 14.953271028037381
F1 Score: 22.535211267605632
TP: 48
FP: 57
TN: 553
FN: 273
```



Alpha = 0.08
Iterations = 10

	Real		
Prediction		Positive (1)	Negative (0)
	Positive (1)	TP: 48	FP: 273
	Negative (0)	F.N: 57	TN: 553

Neural Network

Dataset: Heart

StandardScaler

Description: Description:

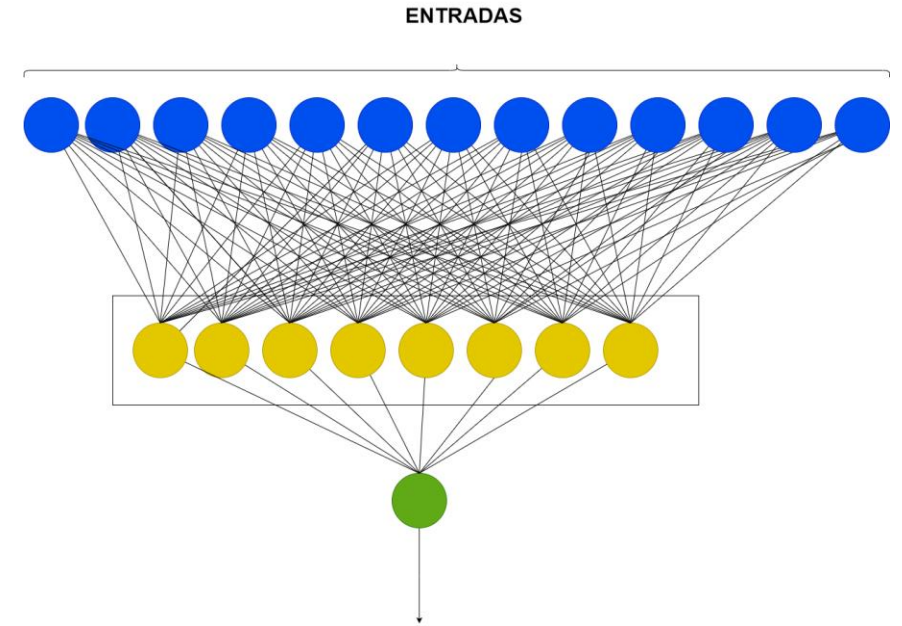
age
sex
chest pain type (4 values)
resting blood pressure
serum cholestoral in mg/dl
fasting blood sugar > 120 mg/dl
resting electrocardiographic results (values 0,1,2)
maximum heart rate achieved
exercise induced angina
oldpeak = ST depression induced by exercise relative to rest
the slope of the peak exercise ST segment
number of major vessels (0-3) colored by flourosopy
thal: 0 = normal; 1 = fixed defect; 2 = reversable defect

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	age	sex	cp	trtbps	chol	fbs	restecg	thalachh	exng	oldpeak	slp	caa	thall	output
2	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
3	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
4	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
5	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
6	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1
7	57	1	0	140	192	0	1	148	0	0.4	1	0	1	1
8	56	0	1	140	294	0	0	153	0	1.3	1	0	2	1
9	44	1	1	120	263	0	1	173	0	0	2	0	3	1
10	52	1	2	172	199	1	1	162	0	0.5	2	0	3	1
11	57	1	2	150	168	0	1	174	0	1.6	2	0	2	1
12	54	1	0	140	239	0	1	160	0	1.2	2	0	2	1
13	48	0	2	130	275	0	1	139	0	0.2	2	0	2	1
14	49	1	1	130	266	0	1	171	0	0.6	2	0	2	1
15	64	1	3	110	211	0	0	144	1	1.8	1	0	2	1
16	58	0	3	150	283	1	0	162	0	1	2	0	2	1
17	50	0	2	120	219	0	1	158	0	1.6	1	0	2	1

$$\frac{\partial C}{\partial w^L} = \frac{\partial C}{\partial a^L} \cdot \frac{\partial a^L}{\partial z^L} \cdot \frac{\partial z^L}{\partial w^L}$$

$$\frac{\partial C}{\partial b^L} = \frac{\partial C}{\partial a^L} \cdot \frac{\partial a^L}{\partial z^L} \cdot \frac{\partial z^L}{\partial b^L}$$

$$\frac{\partial a^L}{\partial z^L} = a^L(z^L) \cdot (1 - a^L(z^L)) \quad \frac{\partial C}{\partial a_j^L} = (a_j^L - y_j) \quad \frac{\partial z^L}{\partial b^L} = 1 \quad \frac{\partial z^L}{\partial w^L} = a_i^{L-1}$$



OPERATION

$$\delta^L = \frac{\partial C}{\partial a^L} \cdot \frac{\partial a^L}{\partial z^L} \cdot \frac{\partial z^L}{\partial w^L}$$

$$\delta^{L-1} = \frac{\partial C}{\partial a^{L-1}} \cdot \frac{\partial a^{L-1}}{\partial z^{L-1}} \cdot \frac{\partial z^{L-1}}{\partial w^{L-1}}$$

$$\delta^{L-2} = \frac{\partial C}{\partial a^{L-2}} \cdot \frac{\partial a^{L-2}}{\partial z^{L-2}} \cdot \frac{\partial z^{L-2}}{\partial w^{L-2}}$$

APLICAMOS LA CHAIN RULE A ESTA COMPOSICION

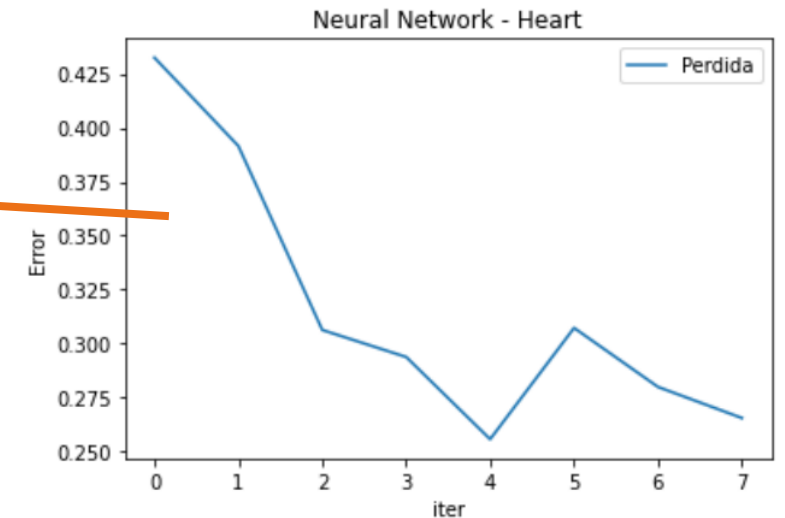
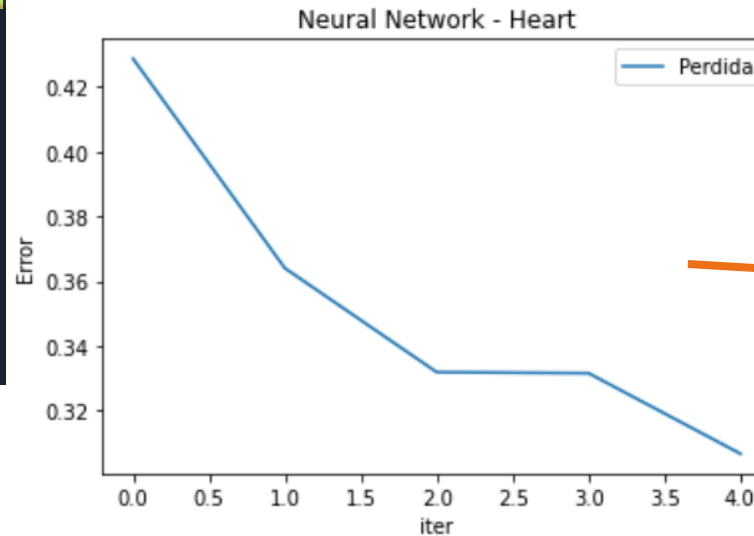
$$C(a^L(W^L a^{L-1}(W^{L-1} a^{L-2} + b^{L-1}) + b^L))$$

RESULTS

```

unlabeled.py ; md5 = 27051571150
Accuracy: 75.40983606557377
Precision: 77.41935483870968
Sensitivity: 75.0
F1 Score: 76.19047619047619
TP: 24
FP: 7
TN: 22
FN: 8
    
```

Alpha = 3
Iterations = 5



More iterations the cost increment

```

Accuracy: 81.9672131147541
Precision: 86.20689655172413
Sensitivity: 78.125
F1 Score: 81.96721311475409
TP: 25
FP: 4
    
```

	Real		
Prediction		Positive (1)	Negative (0)
	Positive (1)	TP: 24	FP: 7
	Negative (0)	F.N: 8	TN: 22

Neural Network

Dataset: Zoo

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	animal_name	hair	feathers	eggs	milk	airborne	aquatic	predator	toothed	backbone	breathes	venomous	fins	legs	tail	domestic	catsize	class_type
2	aardvark		1	0	0	1	0	0	1	1	1	1	0	0	4	0	0	1
3	antelope		1	0	0	1	0	0	0	1	1	1	0	0	4	1	0	1
4	bass		0	0	1	0	0	1	1	1	1	0	0	1	0	1	0	4
5	bear		1	0	0	1	0	0	1	1	1	1	0	0	4	0	0	1
6	boar		1	0	0	1	0	0	1	1	1	1	0	0	4	1	0	1
7	buffalo		1	0	0	1	0	0	0	1	1	1	0	0	4	1	0	1
8	calf		1	0	0	1	0	0	0	1	1	1	0	0	4	1	1	1
9	carp		0	0	1	0	0	1	0	1	1	0	0	1	0	1	1	4
10	catfish		0	0	1	0	0	1	1	1	1	0	0	1	0	1	0	4
11	cavy		1	0	0	1	0	0	0	1	1	1	0	0	4	0	1	1
12	cheetah		1	0	0	1	0	0	1	1	1	1	0	0	4	1	0	1
13	chicken		0	1	1	0	1	0	0	1	1	1	0	0	2	1	1	2
14	chub		0	0	1	0	0	1	1	1	1	0	0	1	0	1	0	4
15	clam		0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	7
16	crab		0	0	1	0	0											
17	crayfish		0	0	1	0	0											
18	crow		0	1	1	0	1											
19	deer		1	0	0	1	0											
20	dogfish		0	0	1	0	0											
21	dolphin		0	0	0	1	0											
22	dove		0	1	1	0	1											

One Hot Encoding
Estándar Scaler

	A	B	C	D	E	F	G	H	I
1	Class_Numb	Number_Of	Class_Type	Animal_Names					
2	1	41	Mammal	aardvark, antelope, bear, boar, buffalo, calf, cavy, cheetah, deer, dolphin, ele					
3	2	20	Bird	chicken, crow, dove, duck, flamingo, gull, hawk, kiwi, lark, ostrich, parakeet, p					
4	3	5	Reptile	pitvip, seasnake, slowworm, tortoise, tuatara					
5	4	13	Fish	bass, carp, catfish, chub, dogfish, haddock, herring, pike, piranha, seahorse, s					
6	5	4	Amphibian	frog, frog, newt, toad					
7	6	8	Bug	flea, gnat, honeybee, housefly, ladybird, moth, termite, wasp					
8	7	10	Invertebrate	clam, crab, crayfish, lobster, octopus, scorpion, seawasp, slug, starfish, worm					



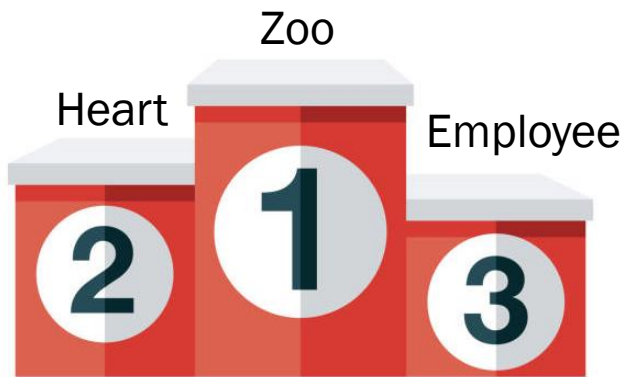
404. That's an error.

The requested URL /doesntexist was not found on this server. That's all we know.

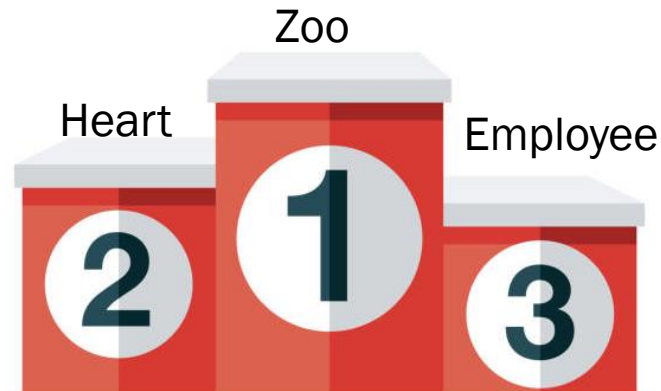


CONCLUSION

- **Logistic Regression was the best algorithm in the zoo dataset**
- An iterations increment can be a problem in the final cost (heart – logistic regression)
- Binary Logistic Regression is not a good algorithm for different targets
- Evaluation metrics are a good tool to evaluate performance
- It is very important to analyze the independent variables and transform the range or use encoding
- Be careful when use a big number of k neighbors in knn algorithm



Logistic Regression



KNN

